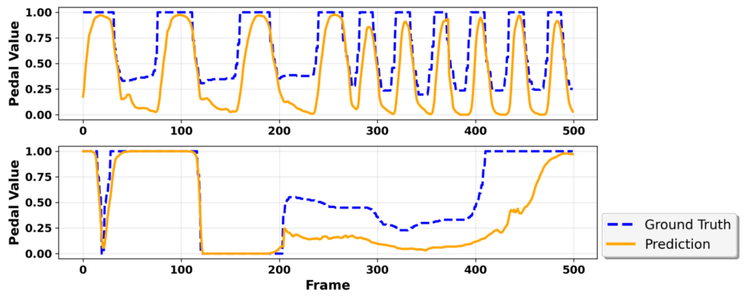
EVALUATING HIGH-RESOLUTION PIANO SUSTAIN PEDAL DEPTH ESTIMATION WITH MUSICALLY INFORMED METRICS

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Problem

Conventional frame-level metrics (MSE/MAE/F1) are insufficient to evaluate continuous piano sustain pedal depth estimation, because they miss musically meaningful boundary accuracy and contour similarity.

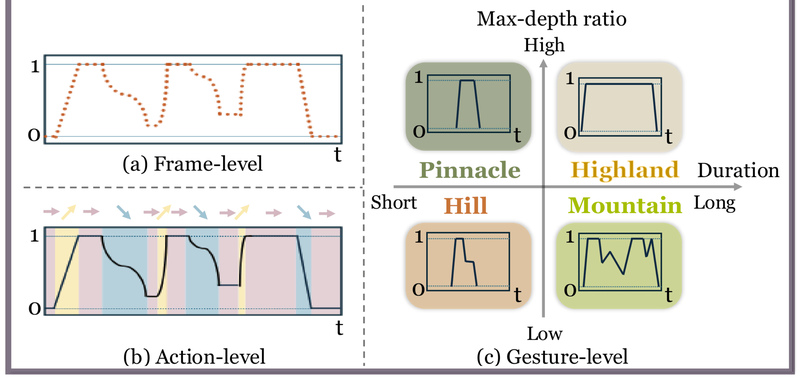


Two examples with large MSE and MAE but correctly following the pedaling pattern. The top one has MSE 0.0983 and MAE 0.2425. The bottom one has MSE 0.0866 and MAE 0.2053.

Main idea

We evaluate sustain pedal depth estimation at three complementary levels:

- **Frame level:** per-frame fidelity
- **Action level:** press / hold / release timing and alignment
- **Gesture level:** contour similarity of complete pedal cycles



Frame-level. AUDIO+MIDI gives the best overall frame-wise performance, but frame-level gains alone do not fully reveal musically relevant improvements.

Model	Binary (P↑/R↑/F1↑)			4-Class (P↑/R↑/F1↑)			MSE ↓	MAE ↓
[11]	0.8975	0.8971	0.8973	0.6849	0.6971	0.6863	0.0425	0.1339
AUDIO (BINARY)	0.8946	0.8944	0.8945	0.6016	0.6627	0.6166	0.0582	0.1502
AUDIO	0.9043	0.9037	0.9039	0.7013	0.7153	0.7045	0.0416	0.1237
AUDIO+MIDI	0.9379	0.9370	0.9372	0.7529	0.7661	0.7546	0.0280	0.0986

Controlled comparison

- **AUDIO (BINARY):** binary target training
- **AUDIO:** continuous regression baseline
- **AUDIO+MIDI:** continuous regression + fused MIDI stream

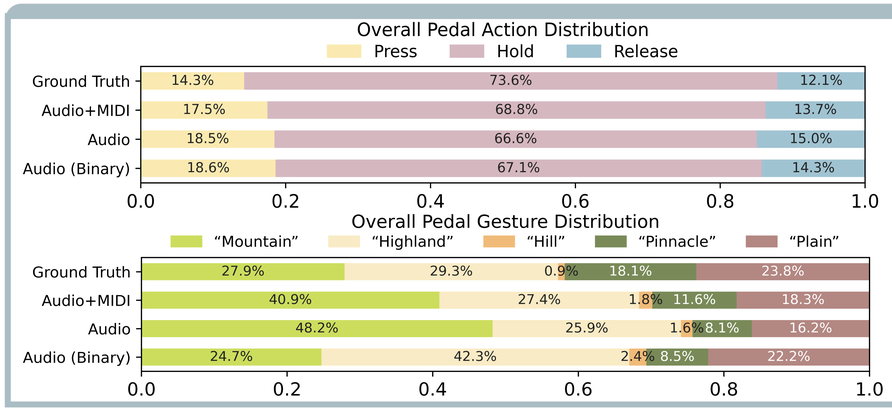
All three variants share the same Transformer-based architecture and training setup.

Action-level. Continuous-valued estimation improves press/release modeling, and AUDIO+MIDI yields the strongest action-level F1.

	Press			*Hold*			*Release*			*Overall*	
Model	P ↑	R ↑	F1 ↑	P ↑	R ↑	F1 ↑	P ↑	R ↑	F1 ↑	F1 (Macro) ↑	F1 (Weighted) ↑
AUDIO (BINARY)	0.5068	0.6605	0.5739	0.8734	0.7961	0.8330	0.5380	0.6347	0.5823	0.6629	0.7655
AUDIO	0.5383	0.6959	0.6070	0.8877	0.8028	0.8431	0.5544	0.6850	0.6128	0.6876	0.7815
AUDIO+MIDI	0.6325	0.7745	0.6964	0.9171	0.8568	0.8859	0.6807	0.7721	0.7235	0.7686	0.8392

Gesture-level. AUDIO+MIDI best matches gesture shapes and distributions, especially for short rapid gestures such as *pinnacle* and *hill*.

	Mountain		*Highland*		*Hill*		*Pinnacle*		*Plain*		Weighted	
Model	5-pts ↓	Fourier ↓	5-pts ↓	Fourier ↓	5-pts ↓	Fourier ↓	5-pts ↓	Fourier ↓	5-pts ↓	Fourier ↓	5-pts ↓	Fourier ↓
AUDIO (BINARY)	0.0863	0.0544	0.1010	0.0207	0.1163	0.0761	0.1136	0.0657	0.1395	0.0512	0.1085	0.0460
AUDIO	0.0689	0.0284	0.0880	0.0146	0.0899	0.0521	0.0941	0.0503	0.1332	0.0471	0.0946	0.0329
AUDIO+MIDI	0.0457	0.0273	0.0441	0.0116	0.0511	0.0358	0.0518	0.0291	0.0737	0.0247	0.0530	0.0225



Take-home messages

- Frame-level metrics alone are insufficient to evaluate continuous pedal depth estimation.
- Continuous-valued prediction is crucial for capturing actions and gesture shapes.
- MIDI provides useful structural priors for better pedal contour modeling.